

RI.UVT.RO Redesign

Week 4 Concept Guide

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Focus: Explanation of the backend and API concepts referenced in the Week 4 summaries: CPTs, taxonomies, ACF, REST endpoints, and field contracts.

Technology Stack: WordPress CMS + REST API + React + Vite +
TypeScript
Team Q-UW9

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1. Document Purpose

This document explains the backend and API concepts that appear in the Week 4 summaries. Its goal is to make terms like CPT, taxonomy, ACF, REST endpoint, field contract, and headless CMS understandable for team members who are not primarily WordPress developers.

2. Headless CMS

2.1 Meaning

A **headless CMS** is a content management system where the backend content editor and the frontend website are separated. WordPress still stores and manages the content, but React is responsible for displaying that content to users.

Simple version: WordPress becomes the content database and admin panel. React becomes the public-facing website.

2.2 How It Applies to RI.UVT.RO

In this project, WordPress is used to define structured content such as calls, programmes, resources, and stories. React requests that content through the WordPress REST API and renders it through reusable components, templates, and routes.

2.3 Why This Matters

- Editors can manage content in WordPress without touching React code.
- Developers can build a modern frontend with React and Vite.
- Content can be reused across pages because it is exposed through API endpoints.
- The structure of the backend must be stable before the frontend API helpers can be reliable.

3. CPT: Custom Post Type

3.1 Meaning

A **Custom Post Type**, usually shortened to **CPT**, is a custom content category in WordPress. WordPress has normal posts and pages by default, but a project can define additional content types when the website needs more specific structures.

Simple version: A CPT is a new kind of WordPress content item, separate from normal posts and pages.

3.2 Project Examples

The Week 4 backend model uses four main CPTs:

CPT	Purpose
calls	Stores calls, opportunities, application announcements, deadlines, eligibility, application steps, and required documents.
programmes	Stores study or mobility programme information such as duration, language, partner institution, and application deadline.
resources	Stores downloadable or reference materials such as PDFs, DOCX files, XLS files, or useful links.
stories	Stores testimonials, student stories, author details, dates, and pull quotes.

3.3 Why CPTs Matter for React

React needs predictable data sources. If calls, programmes, resources, and stories are all separate CPTs, the frontend can request them from separate endpoints and display them using the correct components.

```
/wp-json/wp/v2/calls  
/wp-json/wp/v2/programmes  
/wp-json/wp/v2/resources  
/wp-json/wp/v2/stories
```

4. Taxonomies

4.1 Meaning

A **taxonomy** is a classification system. In WordPress, taxonomies group content items by shared categories, audiences, topics, years, or families.

Simple version: A taxonomy is a filtering and grouping system for content.

4.2 Project Examples

The Week 4 summary mentions filters such as:

- **audience:** groups content by target user, such as international students, partners, or staff.
- **academic-year:** groups calls or resources by academic year.
- **programme-family:** groups programmes by programme type or family.
- **content-topic:** groups resources and stories by subject area.

4.3 Why Taxonomies Matter for the API

Taxonomies make filtered API requests possible. For example, React can request only resources for a specific audience or only calls for a specific academic year.

```
getCalls({ audience: "international-students" })
getResources({ contentTopic: "admissions" })
getProgrammes({ programmeFamily: "erasmus" })
```

Without taxonomies attached to CPTs, those filters would not work reliably. This is why the Week 4 summary treats taxonomy registration and attachment as a major backend requirement.

5. ACF: Advanced Custom Fields

5.1 Meaning

ACF stands for **Advanced Custom Fields**. It is a WordPress plugin that lets developers add custom fields to content types. These fields make WordPress content structured instead of being only one large text editor.

Simple version: ACF defines the exact input fields editors fill in for each content type.

5.2 Project Examples

A normal WordPress page might only have a title and body content. A structured RI.UVT.RO call needs more fields:

- deadline;
- eligibility;
- application steps;
- documents;
- audience notes.

A programme might need:

- application deadline;
- duration;

- language;
- partner institution.

5.3 Why ACF Matters for React

React needs stable field names. If the frontend expects a field named `deadline`, but WordPress exposes it as `application_deadline`, the API helper or page component can break. This is why the Week 4 summary emphasizes a frozen field-name contract.

Implementation principle: ACF field names are part of the API contract. Once React starts consuming them, they should not change casually.

6. REST API Endpoint

6.1 Meaning

A **REST API endpoint** is a URL that returns data from WordPress in a machine-readable format, usually JSON. React calls these endpoints to receive content.

Simple version: An endpoint is the address React uses to ask WordPress for content.

6.2 Project Examples

```
/wp-json/wp/v2/pages  
/wp-json/wp/v2/calls  
/wp-json/wp/v2/programmes  
/wp-json/wp/v2/resources  
/wp-json/wp/v2/stories
```

6.3 Why Endpoint Verification Is Pending

The Week 4 summary marks endpoint verification as pending because the CPTUI configuration and ACF field groups still need to be imported or synced inside WordPress admin. The repository can contain the correct configuration files, but the running WordPress site will not expose the final API until those settings are active in WordPress.

7. CPTUI

7.1 Meaning

CPTUI refers to the Custom Post Type UI plugin. It allows developers to register custom post types and taxonomies through WordPress configuration instead of writing everything manually in PHP.

Simple version: CPTUI is the tool used to define CPTs and taxonomies in WordPress.

7.2 Why CPTUI Import Is Pending

The repository contains export files such as `cptui-export.json` and `cptui-taxonomies.json`. However, these still need to be imported into the active WordPress admin so that the local WordPress instance actually registers the CPTs and taxonomies.

8. Field Name Contract

8.1 Meaning

A **field name contract** is the agreed list of exact field names that WordPress exposes and React consumes. It connects backend structure to frontend code.

Example problem: If the backend calls a field `file_url`, but the frontend expects `download_url`, the resource download component may receive no usable link.

8.2 Why It Matters

The API helpers, React templates, and page components depend on consistent field names. Freezing the field contract means Role 5 can safely write API helpers without guessing what the backend will return.

9. How the Concepts Work Together

Concept	Role in the system
Headless CMS	Separates WordPress content management from the React frontend.
CPT	Defines each major content type, such as calls, programmes, resources, and stories.
Taxonomy	Allows content to be grouped and filtered by audience, topic, academic year, or programme family.
ACF	Adds structured fields to each CPT or page section.

REST endpoint	Exposes WordPress content so React can fetch it.
API helper	A frontend function that wraps endpoint requests into reusable code.
Field contract	Keeps backend field names and frontend expectations aligned.

10. Practical Summary

The Week 4 summaries show that the backend structure is mostly defined in files, but it still needs to be activated and tested inside WordPress. After CPTUI import, ACF sync, and test entries are complete, Role 5 can verify endpoints and connect React pages to real WordPress content.